Exercise Sheet 7

The problems use complex finite-dimensional Lie algebras. Matrix units satisfy $E_{ij}E_{kl} = \delta_{jk}E_{il}$, and κ is the Killing form. In $\mathfrak{sl}_2(\mathbb{C})$ use

$$h = E_{11} - E_{22}, \qquad e = E_{12}, \qquad f = E_{21}.$$

Of the theorems imported in Lecture 7, only the self-centraliser theorem is used, in Exercise 7.2(iii). The diagonal centralisers are proved directly in Exercise 7.1. The sheet also uses the Killing-form results of Lecture 4.

Exercise 7.1 (The diagonal centraliser in $\mathfrak{sl}_n$). Let $n \geq 2$, let $\mathfrak{g} = \mathfrak{sl}_n(\mathbb{C})$, and let $\mathfrak{h} \subseteq \mathfrak{g}$ be the traceless diagonal matrices.

(i) For $H = \operatorname{diag}(a_1, \ldots, a_n)$, prove $[H, X]_{ij} = (a_i - a_j)X_{ij}$. Deduce that $\mathfrak{h}$ is toral.

(ii) Find $H_0 \in \mathfrak{h}$ with $C_{\mathfrak{g}}(H_0) = \mathfrak{h}$. Deduce $C_{\mathfrak{g}}(\mathfrak{h}) = \mathfrak{h}$ directly.

(iii) Prove that $\mathfrak{h}$ is a Cartan subalgebra. Find its roots and root spaces, and check the dimension of $\mathfrak{g}$.

Exercise 7.2 (Maximal abelian subalgebras). A subalgebra is *maximal abelian* if it is abelian and not properly contained in an abelian subalgebra.

(i) Prove that $\mathbb{C}e \subseteq \mathfrak{sl}_2(\mathbb{C})$ is maximal abelian but contains no nonzero semisimple element.

(ii) In $\mathfrak{g} = \mathfrak{sl}_3(\mathbb{C})$, prove that $\mathfrak{a} = \operatorname{span}\{E_{12}, E_{13}\}$ is maximal abelian and contains no nonzero semisimple element.

(iii) Prove that every Cartan subalgebra of a semisimple $\mathfrak{g}$ is maximal abelian.

(iv) In $\mathfrak{sl}_4(\mathbb{C})$, prove that $\mathfrak{n} = \operatorname{span}\{E_{13}, E_{14}, E_{23}, E_{24}\}$ is maximal abelian of dimension 4 and contains no nonzero semisimple element. Compare it with the diagonal Cartan subalgebra.

Exercise 7.3 (The Killing form of $\mathfrak{sl}_2$ on root spaces). In $\mathfrak{sl}_2(\mathbb{C})$ use the basis (h, e, f) fixed at the start of the sheet. Let $\mathfrak{h} = \mathbb{C}h$.

(i) Prove that $\mathfrak{h}$ is a Cartan subalgebra with roots $\pm\alpha$, where $\alpha(h) = 2$.

(ii) Verify the Killing values $\kappa(h, h) = 8$, $\kappa(e, f) = 4$ and $\kappa(e, e) = 0$ from Lecture 4, using the matrices of $\operatorname{ad} h$, $\operatorname{ad} e$ and $\operatorname{ad} f$ in this basis.

(iii) Find t_α, and check $[e, f] = \kappa(e, f)\, t_\alpha$.

Exercise 7.4 (A second Cartan subalgebra of $\mathfrak{sl}_2$). Let $x = e + f \in \mathfrak{sl}_2(\mathbb{C})$, and let

$$P = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

(i) Show that the map $\varphi\colon \mathfrak{sl}_2(\mathbb{C}) \to \mathfrak{sl}_2(\mathbb{C})$, $\varphi(X) = PXP^{-1}$, is an automorphism with $\varphi(h) = x$.

(ii) Deduce that $\mathbb{C}x$ is a Cartan subalgebra. Find its root vectors $\varphi(e)$ and $\varphi(f)$, and check one eigenvalue equation directly.

(iii) Find the roots of $\mathbb{C}x$ and the value $\kappa(x, x)$.

Exercise 7.5 (A toral subalgebra that is not maximal). In $\mathfrak{g} = \mathfrak{sl}_3(\mathbb{C})$ let $T = \operatorname{diag}(1, 1, -2)$ and $\mathfrak{t} = \mathbb{C}T$.

(i) Show that $\mathfrak{t}$ is toral but not a Cartan subalgebra.

(ii) Find the eigenspaces of $\operatorname{ad} T$ and the centraliser $C_{\mathfrak{g}}(\mathfrak{t})$.

(iii) Let $\mathfrak{s}$ be any toral subalgebra of a semisimple Lie algebra $\mathfrak{g}$, with weight spaces $\mathfrak{g}^{\mathfrak{s}}_{\lambda}$ for $\lambda \in \mathfrak{s}^*$. Prove that $\kappa(\mathfrak{g}^{\mathfrak{s}}_{\lambda}, \mathfrak{g}^{\mathfrak{s}}_{\mu}) = 0$ unless $\lambda + \mu = 0$. Prove also that κ pairs $\mathfrak{g}^{\mathfrak{s}}_{\lambda}$ and $\mathfrak{g}^{\mathfrak{s}}_{-\lambda}$ nondegenerately.

(iv) Which conclusions of Lecture 7 fail for $\mathfrak{t}$?

Exercise 7.6 (The Killing form on the roots of $\mathfrak{sl}_3$). Use the notation of the worked example in Lecture 7. For $\lambda, \mu \in \mathfrak{h}^*$, put

$$(\lambda, \mu) = \kappa(t_\lambda, t_\mu).$$

(i) Verify the lecture's values of t_{α_1}, t_{α_2} and $t_{\alpha_1+\alpha_2}$ from their defining pairings.

(ii) Compute the matrix $\big((\alpha_i, \alpha_j)\big)_{i,j=1,2}$ and (α, α) for every root. Deduce that α_1 and α_2 form a basis of $\mathfrak{h}^*$.

(iii) Compute $[E_{23}, E_{32}]$, and deduce $\kappa(E_{23}, E_{32})$ from the bracket formula of Lecture 7.